

Grade 11 Mathematics

Equations and Inequalities & Exponents and Surds

Task: Task

Term: Term 2

Time: 90 minutes

Total Marks: 50

QUESTION 1

1.1 Solve for x : $(x-3)(x+5) \leq 0$ (4)

1.2 Answer the following questions about the nature of the roots of quadratic equations.

1.2.1 Without solving the equation, determine the nature of the roots of $2x^2 - 3x + 5 = 0$. (2)

1.2.2 Determine the value(s) of k for which the equation $x^2 + (k+1)x + 9 = 0$ has equal roots. (3)

1.3 Solve simultaneously for x and y : $2x = y + 3$ and $x^2 + y^2 = 3 + xy$ (6)

1.4 Consider the expressions $(2x-1)^2$ and $(x+4)(2x-1)$.

1.4.1 Expand $(x+4)(2x-1)$. (1)

1.4.2 Hence, or otherwise, determine all real values of x for which $(2x-1)^2 \leq (x+4)(2x-1)$. (4)

1.5 Given: $2x - 3 = \frac{7}{x}$, where $x \neq 0$

1.5.1 Solve for x in the given equation. (correct to TWO decimal places) (4)

1.5.2 Hence, or otherwise, solve for x if $2(x+1)^2 - 3(x+1) - 7 = 0$. (correct to TWO decimal places) (1)

[25]

QUESTION 2

2.1 Rectangle ABCD has length $AB = 3\sqrt{2}$ cm and breadth $BC = \sqrt{8}$ cm.

2.1.1 Calculate the area of rectangle ABCD. Leave your answer in simplest form. (2)

2.1.2 Calculate the length of the diagonal AC. Leave your answer in simplest surd form. (2)

2.2 Without using a calculator, simplify each of the following expressions completely. Show all working.

2.2.1
$$\frac{\sqrt[3]{64} \cdot 16^{\frac{-3}{4}}}{\sqrt{\frac{1}{9}}}$$
 (3)

2.2.2
$$\frac{\sqrt{27} \cdot 9^{\frac{-1}{4}}}{\sqrt[5]{243}^{-2}}$$
 (3)

2.3 Solve for x : $3^{2x+1} = 8 \cdot 3^x + 3$ (5)

2.4 Consider the expression $E = \frac{\sqrt{x^2 - 4} \cdot (x + 2)^{\frac{5}{2}}}{(x - 2)^{\frac{1}{2}}}$, where $x > 2$.

2.4.1 Factorise $x^2 - 4$ fully. (1)

2.4.2 Hence, or otherwise, determine the value of x for which $E = 125$. (4)

2.5 Given the expression $P = \sqrt{x + \sqrt{2x - 1}} \cdot \sqrt{x - \sqrt{2x - 1}}$, where $x > 1$.

2.5.1 Simplify P fully. (3)

2.5.2 Determine the value of x for which (2)

$$\sqrt{x+\sqrt{2x-1}} \cdot \sqrt{x-\sqrt{2x-1}} = 12.$$

[25]

TOTAL [50]

MARKING GUIDELINE

QUESTION 1		
1.1	Critical values: $x=3$ or $x=-5$	✓ critical values
	Sign diagram: $+$ for $x < -5$, $-$ for $-5 < x < 3$, $+$ for $x > 3$	✓ sign diagram or sketch
	$-5 \leq x \leq 3$ or $x \in [-5; 3]$	✓ ✓ both bounds (4)
1.2.1	$\Delta = (-3)^2 - 4(2)(5)$	✓ substitution
	$\Delta = 9 - 40 = -31$, so $\Delta < 0$	
	The roots are non-real because $\Delta = -31 < 0$.	✓ conclusion (2)
1.2.2	$\Delta = (k+1)^2 - 4(1)(9)$	✓ substitution
	$k^2 + 2k - 35 = 0$ (setting $\Delta = 0$)	✓ condition $\Delta = 0$
	$(k-5)(k+7) = 0$	
	$k = 5$ or $k = -7$	✓ answer (3)
1.3	$y = 2x - 3$	✓ y as subject
	$x^2 + (2x-3)^2 = 3 + x(2x-3)$	✓ substitution

	$3x^2 - 9x + 6 = 0$, i.e. $x^2 - 3x + 2 = 0$	✓ standard form
	$(x-1)(x-2) = 0$	✓ factors
	$x=1, y=-1$ or $x=2, y=1$	✓ ✓ both pairs of values (6)
1.4.1	$2x^2 + 7x - 4$	✓ answer (1)
1.4.2	$(2x-1)^2 - (x+4)(2x-1) \leq 0$, i.e. $2x^2 - 11x + 5 \leq 0$	✓ standard form
	$(2x-1)(x-5) \leq 0$	✓ factors
	$\frac{1}{2} \leq x \leq 5$	✓ ✓ answer, both parts (4)
1.5.1	$2x^2 - 3x - 7 = 0$	✓ standard form
	$x = \frac{-(-3) \pm \sqrt{(-3)^2 - 4(2)(-7)}}{2(2)}$	✓ substitution
	$x=2,77$ or $x=-1,27$	✓ ✓ both answers (4)
1.5.2	$x+1=2,77$ or $x+1=-1,27$	
	$x=1,77$ or $x=-2,27$	✓ both answers (1)

QUESTION 2		
2.1.1	Area = $3\sqrt{2} \times \sqrt{8}$	✓ substitution
	Area = 12 cm^2	✓ answer (2)
2.1.2	$AC^2 = (3\sqrt{2})^2 + (\sqrt{8})^2 = 18 + 8$	✓ substitution into Pythagoras
	$AC = \sqrt{26} \text{ cm}$	✓ answer (2)
2.2.1	$\frac{(2^6)^{\frac{1}{3}} \cdot (2^4)^{\frac{-3}{4}}}{(3^{-2})^{\frac{1}{2}}}$	✓ convert to prime-base exponent form
	$\frac{2^2 \cdot 2^{-3}}{3^{-1}} = \frac{4 \cdot \frac{1}{8}}{\frac{1}{3}}$	✓ simplify each term
	$\frac{3}{2}$	✓ answer (3)
2.2.2	$\frac{3^{\frac{3}{2}} \cdot 3^{\frac{-1}{2}}}{3^{-2}}$	✓ convert to prime-base exponent form
	$3^{\frac{3}{2} - \frac{1}{2} + 2} = 3^3$	✓ simplify exponents
	27	✓ answer (3)

2.3	Let $k=3^x$: $3 \cdot 3^{2x} = 8 \cdot 3^x + 3$ becomes $3k^2 = 8k + 3$	✓ substitution
	$3k^2 - 8k - 3 = 0$	✓ standard form
	$(3k+1)(k-3) = 0$	✓ factors
	$3^x = \frac{-1}{3}$ (no solution, since $3^x > 0$) or $3^x = 3$	✓ values of k with rejection
	$x = 1$	✓ answer (5)
2.4.1	$(x-2)(x+2)$	✓ factors (1)
2.4.2	$\sqrt{x^2 - 4} = (x-2)^{\frac{1}{2}}(x+2)^{\frac{1}{2}}$	✓ surd as fractional exponents
	$E = \frac{(x-2)^{\frac{1}{2}}(x+2)^{\frac{1}{2}}(x+2)^{\frac{5}{2}}}{(x-2)^{\frac{1}{2}}} = (x+2)^3$	✓ combine exponents
	$(x+2)^3 = 125$ so $x+2 = 5$	✓ cube root
	$x = 3$	✓ answer (4)
2.5.1	$P = \sqrt{(x+\sqrt{2x-1})(x-\sqrt{2x-1})}$	✓ single surd
	$= \sqrt{x^2 - (2x-1)} = \sqrt{x^2 - 2x + 1}$	✓ difference of two squares
	$= \sqrt{(x-1)^2} = x-1$ (since $x > 1$)	✓ factorise to answer

	$P = x - 1$	(3)
2.5.2	$x - 1 = 12$	✓ use simplified form
	$x = 13$	✓ answer (2)

TOTAL [50]

COGNITIVE LEVEL ANALYSIS GRID

QUESTION	Knowledge	Routine	Complex	Problem-Solving	Total
QUESTION 1	5	9	7	4	25
QUESTION 2	5	9	7	4	25
TOTAL	10	18	14	8	50
FET TARGET %	20%	35%	30%	15%	—
THIS TASK %	20%	36%	28%	16%	100%

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$A = P(1 + ni)$$

$$A = P(1 - ni)$$

$$A = P(1 - i)^n$$

$$A = P(1 + i)^n$$

$$T_n = a + (n-1)d$$

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(r^n - 1)}{r - 1} ; r \neq 1$$

$$S_\infty = \frac{a}{1 - r} ; -1 < r < 1$$

$$F = \frac{x[(1+i)^n - 1]}{i}$$

$$P = \frac{x[1 - (1+i)^{-n}]}{i}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$M\left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2}\right)$$

$$y = mx + c$$

$$y - y_1 = m(x - x_1)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \tan \theta$$

$$(x-a)^2 + (y-b)^2 = r^2$$

$$\text{In } \triangle ABC: \quad \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

$$\text{area } \triangle ABC = \frac{1}{2}ab \cdot \sin C$$

$$\sin(\alpha + \beta) = \sin \alpha \cdot \cos \beta + \cos \alpha \cdot \sin \beta$$

$$\sin(\alpha - \beta) = \sin \alpha \cdot \cos \beta - \cos \alpha \cdot \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cdot \cos \beta - \sin \alpha \cdot \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cdot \cos \beta + \sin \alpha \cdot \sin \beta$$

$$\cos 2\alpha = \begin{cases} \cos^2 \alpha - \sin^2 \alpha \\ 1 - 2 \sin^2 \alpha \\ 2 \cos^2 \alpha - 1 \end{cases}$$

$$\sin 2\alpha = 2 \sin \alpha \cdot \cos \alpha$$

$$\bar{x} = \frac{\sum x}{n}$$

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$\hat{y} = a + bx$$

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$